

Scaffold: Autonomous Research on Hierarchical Agent Execution

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Eligibility gate failed. This document is a structured scaffold, not a paper eligible for submission. It contains no qualifying autonomous ML research result. Part II includes explicit human-authorship markers.

Part I — Qualifying ML research result (maximum 4 pages)

Status: missing; do not replace with the workflow system itself.

Required abstract content. The final abstract must identify a concrete ML research result and explain the autonomous system’s end-to-end or decisive contribution. No such result was completed in this run.

Research question and contribution

Placeholder for a future frozen ML question, method, and result. A study of whether hierarchy improves language-agent performance could qualify only after complete matched-budget experiments, objective graders, statistical analysis, and evidence that the autonomous system performed the qualifying research.

Method, evidence, and result

No quantitative result may be inserted manually. Generate tables from the experiment evidence store. Report model versions, prompts, task instances, budgets, seeds, exclusions, statistical intervals, and negative results.

Limitations

Distinguish system-development evidence from the qualifying ML result and disclose human interventions.

Part II — Autonomous system/design (maximum 4 pages)

[HUMAN AUTHORSHIP / REWRITE REQUIRED BEFORE SUBMISSION]

The following subsections contain verified factual notes and a structure for a human author. They are not presented as venue-compliant human-written prose.

System boundary and authority

[HUMAN REWRITE REQUIRED] Verified facts: the package uses the hierarchy Plan, Phase, DIC, EPS, Prompt/Run, and Action; '.ai-workflow/STATE.json' is the canonical authority index; Main, Theory, and Experiment own persistent workstreams; ordinary phases proceed inside an approved Plan; periodic and material events are human boundaries.

Research-loop autonomy

[HUMAN REWRITE REQUIRED] Describe exactly which hypothesis generation, literature search, implementation, experiment selection, debugging, analysis, and writing steps were autonomous for the qualifying Part I result. The current reconstruction cannot supply this because Part I is absent.

Verification, intervention, and audit trail

[HUMAN REWRITE REQUIRED] Verified system facts: DICs bind bounded claims to evidence; verification returns accept/repair/replan/escalate; DIC/EPS/prompt-run/Action lineage is package-resident; material objective, scientific, protocol, release, paid-compute, and credential changes stop autonomy.

Human contribution accounting

[HUMAN REWRITE REQUIRED] Provide a chronological intervention log and identify which decisions were made by humans. Do not infer human authorship from an AI-generated narrative.

Part III — Reflections (maximum 1 page)

[HUMAN AUTHORSHIP / REWRITE REQUIRED BEFORE SUBMISSION]

A human reflection should discuss where autonomy was scientifically useful, where the system optimized the wrong proxy, how verification failed, what evidence forced replanning, and which interventions remained irreducibly human. The current run suggests one provisional lesson only: persistent contracts and evidence lineage make intervention points easier to audit, but they do not create a qualifying result or guarantee verifier soundness.

References